

CS2422 Assembly Language & System Programming

September 16, 2003

Today's Topics (1/2)

- Data representation – What are stored in the main memory?
 - Binary, Decimal, Hexadecimal...etc.
 - Units: Byte, Word, Double-word...etc.
 - Memory Addresses.
 - Characters Strings.
 - Big or Little Endian.

Today's Topics (2/2)

- How the CPU executes a program.
- Where we can store the data.
 - Registers
 - Memory
 - Constants

Study Guide

- Textbook: Sections 1.3, 2.1 (except 2.1.2.2), 2.2, 2.3.
- Read (in depth):
 - 2.1.1 and 2.1.2.1
 - 2.2.1 and 2.2.2
 - 2.3.1 and 2.3.2.1
- Browse the rest (like reading news stories).

Data Representations

- Consider this simple view: The whole program is stored in main memory.
 - Including program instructions (code) and data.
 - CPU loads the instructions and data from memory for execution.
 - Don't worry about the disk for now.

Number Systems

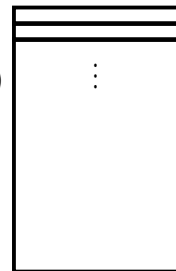
- I assume that you all know:
 - Binary & hexadecimal numbers.
 - Conversion between them and decimal.
 - How to represent negative numbers (in 2's compliment).

Byte? Word?

- The number has limited range.
- 1 Byte = 8 Bits:
 - Binary: 0000 0000 to 1111 1111
 - Hexadecimal: 00 to FF
 - Decimal: 0 to 255
- Word = 2 or 4 bytes, depending on the machine. In 80x86, it means 2 bytes.

Memory Size

- Organized like mailboxes, numbered 0, 1, 2, 3, ..., $2^n - 1$.
 - Each box can hold 8 bits (1 byte)
 - So it is called byte-addressing.



Memory Addresses

- Byte Addressing: each memory location has 8 bits, i.e., one byte.
- If we have only 16 bytes...
 - Then 4 bits are enough for addresses
- What if we have 64K?
- 1M? 1G?

Memory Address

- 16-bit address is enough for up to 64K
- 20-bit for 1M
- 32-bit for 4G
- Most servers need more than 4G!!
That's why we need 64-bit CPUs like Alpha (DEC/Compaq/HP) or Merced (Intel).

Confused Now?

- So, is the memory 8-bit or 32-bit?
- It depends on what you ask:
 - Content or Address?
 - Remember addresses can be stored in memory as well. (They are called pointers in PASCAL and C.)

Character String

- So how are strings like “Hello, World!” are stored in memory?
- ASCII Code!
- Each character is stored as a byte.
- Review: how is “1234” stored in memory?

Integer

- A byte can hold an integer number:
 - between 0 and 255 (unsigned) or
 - between -128 and 127 (2's complement)
- How to store a bigger number?
- Review: how is 1234 stored in memory?

Big or Little Endian?

- Example: 1234 is stored in 2 bytes.
- = 100-1101-0010 in binary
- = 04 D2 in hexadecimal
- Do you store 04 or D2 first?
- Big Endian: 04 first.
- Little Endian: D2 first. ← Intel's choice

Ancient History -- Segment

- IBM PC XT (Intel 8088/8086) is a so-called 16-bit machine.
- Each register has 16 bits.
- $2^{16} = 65536 = 64K$
- But we want to use more memory (640K, 1M)...

Break

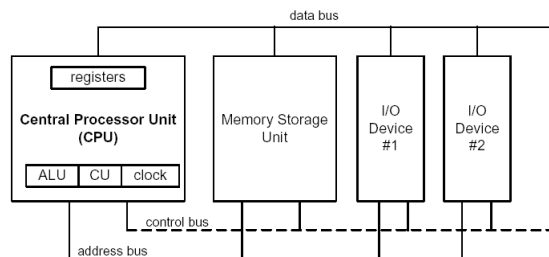
...Continued from Last Lecture

- Reason for Little-Endian?
 - More consistent for variable length (e.g., 2 bytes, 4 bytes, 8 bytes...etc.)

Program Loading & Execution

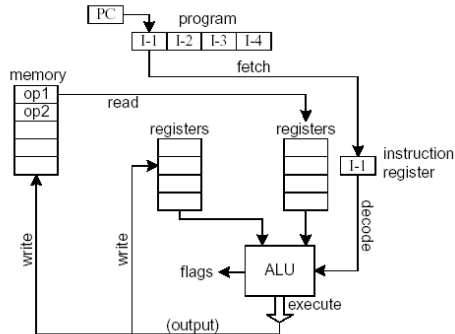
- A simpler view:

Figure 2-1 Block Diagram of a Microcomputer.



- A slightly scarier view:

Figure 2-2 Instruction Execution Cycle.

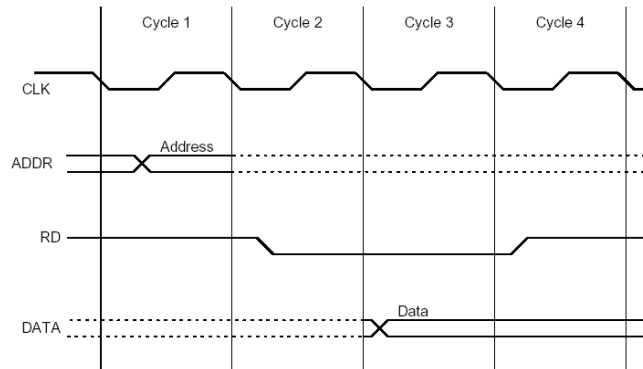


Clock

- Not the wall clock.
- Ever wondering what the CPU clock rate (like 2.0GHz) means?
- Next slide: using the memory read as an example

Figure 2.7 (from Textbook)

Figure 2-7 Memory Read Cycle.

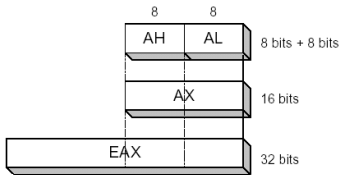


Where are the Data?

- Registers (inside the CPU)
- Memory
- Constant

Registers

- General-Purpose:
 - AX, BX, CX, DX: 16 bits



- Split into H and L parts, 8 bits each.
 - Extended into E?X to become 32-bit register (i.e., EAX, EBX,...etc.).

Convention

- AX: accumulator
- BX: base register
- CX: count register
- DX: data register

Some instructions use them implicitly. (e.g.,
LOOP uses CX with no mentioning.)

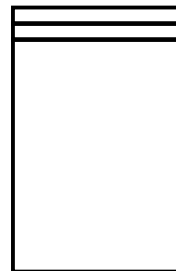
Other Registers

We will explain their meaning when we encounter them later this semester:

- Segment (CS, DS, SS, ES)
- Pointer (IP, SP, BP)
- Index (SI, DI)
- Flags

Memory

- Byte Addressing: each memory location has 8 bits.
- If we have only 16 bytes...
 - 4 bits are enough for an address
- What if we have 64K?
- 1M? 1G?



Memory Address

A Quick Review:

- 16-bit address is enough for up to 64K
- 20-bit for 1M
- 32-bit for 4G

Segment?

- IBM PC XT (Intel 8088/8086) is a so-called 16-bit machine.
- Each register has 16 bits.
- $2^{16} = 65536 = 64K$
- But we want to use more memory (640K, 1M)...

Segment

Segment : Offset

- Segment: one of CS, DS, SS, ES
- Real address = Segment * 16 + Offset
- Overlapping segments. For example:
 $0000:01F0 = 0001:01E0 = 0010:00F0$

Writing the First Program...

- MOV AX, var1
- MOV BX, 1
- ADD AX, BX
- MOV var1, AX

Constant (or Immediate) Data

- Compare:
 - MOV AX, 25
 - MOV AX, var1 ; assume var1 at address 25
- The constant 25 in the first case is called immediate data.
- The second case may be considered as:
 - MOV AX, [25]

Food for Thought

- Example: MOV AX, 25
- But, where is the number stored? In memory?
- A related question: where is the instruction (MOV AX, 25) stored?
- What do we do if we want to move the data in memory address 25 to AX?

Instruction Set Architecture

... the attributes of a [computing] system as seen by the programmer, *i.e.* the conceptual structure and functional behavior, as distinct from the organization of the data flows and controls, the logic design, and the physical implementation.

– Amdahl, Blaaw, and Brooks, 1964

- interface between hardware and low-level software
- standardizes instructions, machine language bit patterns, etc.
- advantage: *different implementations of the same architecture*
- disadvantage: *sometimes prevents using new innovations*

RISC vs. CISC

- RISC: PowerPC, MIPS, DEC Alpha
- CISC: x86/Pentium, AMD, VAX
- Example:

x86 (CISC):

MOV AX, a

ADD AX, b

RISC:

Load R1, a

Load R2, b

Add R2, R1